

Supplementary Note 1: SparseAEH model

1 Gaussian mixture model with spatially-derived full covariance

Let $\mathbf{y} = (y_1, y_2, \dots, y_N)$ correspond to a vector of gene expression over N spots $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$, where each spot $\mathbf{x}_i$ is represented by 2-dimensional vector (x_{i1}, x_{i2}) . The expression matrix $\mathbf{Y}$ consists of G expression vectors $(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_G)$. Assume that there are K underlying patterns for the Gaussian mixture models, and the probability of each expression vector being sampled from the k -th pattern is π_k , such that $\sum_{k=1}^K \pi_k = 1$. A binary matrix $\mathbf{Z}$ indicates the gene g is explained by pattern k if $z_{g,k} = 1$. The expression model can be formulated as the following Gaussian mixture model:

$$\begin{aligned} P(\mathbf{Y}) &= \prod_{g=1}^G P(\mathbf{y}_g) \\ &= \prod_{g=1}^G \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I})) \right\} \end{aligned}$$

where $\boldsymbol{\mu}_k$ denotes the mean vector of the k -th Gaussian pattern and $\boldsymbol{\Sigma}$ denotes the covariance matrix universal for all patterns, realized by the RBF kernel:

$$\Sigma_{i,j} = k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2 \cdot l^2}\right) \quad (1)$$

2 Sparse approximation with block-wise covariance

Partition the N spots into M sets of superspots, $\{S_1, S_2, \dots, S_M\}$ such that every spot belongs to one of the superspots. Let $\{\mathbf{X}_i\}$ be the set of spots belonging to $\{S_i\}$ and

$$\begin{aligned} \boldsymbol{\mu}_{k(i,j,\dots)} &= \boldsymbol{\mu}_k[\mathbf{X}_i, \mathbf{X}_j, \dots] \\ \boldsymbol{\Sigma}_{(i,j,\dots)} &= \boldsymbol{\Sigma}[\mathbf{X}_i, \mathbf{X}_j, \dots; \mathbf{X}_i, \mathbf{X}_j, \dots] \\ \boldsymbol{\Sigma}_{(i,j,\dots),(k,l,\dots)} &= \boldsymbol{\Sigma}[\mathbf{X}_i, \mathbf{X}_j, \dots; \mathbf{X}_k, \mathbf{X}_l, \dots] \end{aligned}$$

Let $\mathbf{y}_{gi}$ be the sub-vector of $\mathbf{y}_g$ corresponding to the spots $\{S_i\}$.

$$\ln p(\mathbf{Y} | \boldsymbol{\mu}, \sigma_k^2, \delta_k) = \sum_{g=1}^G \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I})) \right\} \quad (2)$$

To limit the number of other spots that each spot is correlated with, let $\mathcal{C}_i$ be the set of indices of

superspots that S_i is dependent on. Without loss of generality, we can further assume that $\forall i \in C_j, i < j$.

$$\begin{aligned}
\mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I})) &= \mathcal{N}\left(\bigcap_{i=1}^M \mathbf{y}_{gi} | \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I})\right) \\
&= \prod_{i=1}^M P(\mathbf{y}_{gi} | \mathbf{y}_{g1}, \mathbf{y}_{g2}, \dots, \mathbf{y}_{g(i-1)}, \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I})) \\
&= \mathcal{N}(\mathbf{y}_{g1} | \boldsymbol{\mu}_{k(1)}, \sigma_k^2 \cdot (\boldsymbol{\Sigma}_{(1)(1)} + \delta_k \mathbf{I})) \cdot \\
&\quad \prod_{i=2}^M \mathcal{N}(\mathbf{y}_{gi} | \boldsymbol{\mu}_{k(i)} + \boldsymbol{\Sigma}_{(i)(1, \dots, i-1)} (\boldsymbol{\Sigma}_{(1, \dots, i-1)} + \delta_k \mathbf{I})^{-1} (\mathbf{y}_{g(1, \dots, i-1)} - \boldsymbol{\mu}_{k(1, \dots, i-1)}), \\
&\quad \sigma_k^2 (\boldsymbol{\Sigma}_{(i)(i)} + \delta_k \mathbf{I} - \boldsymbol{\Sigma}_{(i)(1, \dots, i-1)} (\boldsymbol{\Sigma}_{(1, \dots, i-1)} + \delta_k \mathbf{I})^{-1} \boldsymbol{\Sigma}_{(1, \dots, i-1)(i)})) \cdot \\
&\approx \prod_{m=1}^M \mathcal{N}(\mathbf{y}_{gm} | \boldsymbol{\mu}_{k(m)} + \boldsymbol{\Sigma}_{(m)(C_m)} (\boldsymbol{\Sigma}_{(C_m)} + \delta_k \mathbf{I})^{-1} (\mathbf{y}_{g(C_m)} - \boldsymbol{\mu}_{k(C_m)}), \\
&\quad \sigma_k^2 (\boldsymbol{\Sigma}_{(m)} + \delta_k \mathbf{I} - \boldsymbol{\Sigma}_{(m)(C_m)} (\boldsymbol{\Sigma}_{(C_m)} + \delta_k \mathbf{I})^{-1} \boldsymbol{\Sigma}_{(C_m)(m)})) \dots
\end{aligned} \tag{3}$$

Let

$$\begin{aligned}
\boldsymbol{\mu}'_{k(m)} &= \boldsymbol{\mu}_{k(m)} + \boldsymbol{\Sigma}_{(m)(C_m)} (\boldsymbol{\Sigma}_{(C_m)} + \delta_k \mathbf{I})^{-1} (\mathbf{y}_{g(C_m)} - \boldsymbol{\mu}_{k(C_m)}) \\
\boldsymbol{\Sigma}'_{(m)} &= \boldsymbol{\Sigma}_{(m)} + \delta_k \mathbf{I} - \boldsymbol{\Sigma}_{(m)(C_m)} (\boldsymbol{\Sigma}_{(C_m)} + \delta_k \mathbf{I})^{-1} \boldsymbol{\Sigma}_{(C_m)(m)}
\end{aligned}$$

then

$$\begin{aligned}
\mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I})) &\approx \prod_{m=1}^M \mathcal{N}(\mathbf{y}_{gm} | \boldsymbol{\mu}'_{k(m)}, \boldsymbol{\Sigma}'_{(m)}) \\
&= \prod_{m=1}^M (2\pi)^{-\frac{|\mathbf{x}_m|}{2}} \sigma_s^{-|\mathbf{x}_m|} \det(\boldsymbol{\Sigma}'_{(m)})^{-\frac{1}{2}} \cdot \\
&\quad \exp\left(-\frac{1}{2} \sigma_k^{-2} (\mathbf{y}_{gm} - \boldsymbol{\mu}'_{k(m)})^T (\boldsymbol{\Sigma}'_{(m)})^{-1} (\mathbf{y}_{gm} - \boldsymbol{\mu}'_{k(m)})\right) \\
&= (2\pi)^{-\frac{N}{2}} \sigma_s^{-N} \prod_{m=1}^M \det(\boldsymbol{\Sigma}'_{(m)})^{-\frac{1}{2}} \cdot \\
&\quad \exp\left(-\frac{1}{2} \sigma_k^{-2} \sum_{m=1}^M (\mathbf{y}_{gm} - \boldsymbol{\mu}'_{k(m)})^T (\boldsymbol{\Sigma}'_{(m)})^{-1} (\mathbf{y}_{gm} - \boldsymbol{\mu}'_{k(m)})\right)
\end{aligned} \tag{4}$$

3 Estimation with EM algorithm

With the block-wise covariance approximation we stated in Section 2, we are ready to apply **EM algorithm to solve the clustering problem**, in which the major difficulty for high-dimensional case lies in estimating the posterior:

$$\Omega_{gk} = \frac{\pi_k \mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I}))}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_j, \sigma_j^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I}))} \tag{5}$$

Take **derivative w.r.t mean** yields the **local MLE of $\boldsymbol{\mu}_k$** :

$$\frac{\partial \ln p(\mathbf{Y} | \boldsymbol{\mu}, \sigma_k^2, \delta_k)}{\partial \boldsymbol{\mu}_k} = \sum_{g=1}^G \Omega_{gk} (\mathbf{y}_g - \boldsymbol{\mu}_k)^T \sigma^{-2} (\boldsymbol{\Sigma} + \delta_k \mathbf{I})^{-1} \tag{6}$$

$$\boldsymbol{\mu}_k^{new} = \frac{\sum_{g=1}^G \Omega_{gk} \mathbf{y}_g}{\sum_{g=1}^G \Omega_{gk}} \quad (7)$$

To update δ_k and σ_k^2 , we minimize the Frobenius norm

$$\|\sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I}) - \frac{1}{N_k} \sum_{n=1}^N \Omega_{gk} (\mathbf{x}_n - \boldsymbol{\mu}_k^{new})(\mathbf{x}_n - \boldsymbol{\mu}_k^{new})^T\|_F \quad (8)$$

The optimal solution can also be calculated straightforwardly:

$$\sigma_k^{2new} = \frac{\sum_{i,j} M_{ij} \boldsymbol{\Sigma}_{ij} - \sum_{i=1}^N M_{ii} \sum_{i=1}^N \boldsymbol{\Sigma}_{ii}/N}{\sum_{i,j} \boldsymbol{\Sigma}_{ij}^2 - (\sum_{i=1}^N \boldsymbol{\Sigma}_{ii})^2/N} \quad (9)$$

$$\delta_k^{new} = \frac{\sum_{i=1}^N M_{ii} - \sigma_k^{2new} \sum_{i=1}^N \boldsymbol{\Sigma}_{ii}}{N \sigma_k^{2new}} \quad (10)$$

where $M = \frac{1}{N_k} \sum_{n=1}^N \Omega_{gk} (\mathbf{x}_n - \boldsymbol{\mu}_k^{new})(\mathbf{x}_n - \boldsymbol{\mu}_k^{new})^T$

To facilitate easier update in the algorithm, we first derive:

$$\begin{aligned} & \boldsymbol{\Sigma}'_{(m)|\delta=\delta_k} - \boldsymbol{\Sigma}'_{(m)|\delta_k=0} \\ &= \delta_k \mathbf{I} - \boldsymbol{\Sigma}_{(m)(C_m)} (\boldsymbol{\Sigma}_{(C_m)(C_m)} + \delta_k \mathbf{I})^{-1} \boldsymbol{\Sigma}_{(C_m)(m)} + \boldsymbol{\Sigma}_{(m)(C_m)} (\boldsymbol{\Sigma}_{(C_m)(C_m)})^{-1} \boldsymbol{\Sigma}_{(C_m)(m)} \\ &= \delta_k \mathbf{I} - \boldsymbol{\Sigma}_{(m)(C_m)} (\mathbf{P}_m \mathbf{D}_m \mathbf{P}_m^T + \delta_k \mathbf{I})^{-1} \boldsymbol{\Sigma}_{(C_m)(m)} + \boldsymbol{\Sigma}_{(m)(C_m)} (\mathbf{P}_m \mathbf{D}_m \mathbf{P}_m^T)^{-1} \boldsymbol{\Sigma}_{(C_m)(m)} \\ &= \delta_k \mathbf{I} - \boldsymbol{\Sigma}_{(m)(C_m)} (\mathbf{P}_m \mathbf{D}_m \mathbf{P}_m^T + \delta_k \mathbf{I})^{-1} \boldsymbol{\Sigma}_{(C_m)(m)} + \boldsymbol{\Sigma}_{(m)(C_m)} (\mathbf{P}_m \mathbf{D}_m \mathbf{P}_m^T)^{-1} \boldsymbol{\Sigma}_{(C_m)(m)} \\ &= \delta_k \mathbf{I} - \boldsymbol{\Sigma}_{(m)(C_m)} \mathbf{P}_m (\mathbf{D}_m + \delta_k \mathbf{I})^{-1} \mathbf{P}_m^T \boldsymbol{\Sigma}_{(C_m)(m)} + \boldsymbol{\Sigma}_{(m)(C_m)} \mathbf{P}_m \mathbf{D}_m^{-1} \mathbf{P}_m^T \boldsymbol{\Sigma}_{(C_m)(m)} \\ &= \delta_k \mathbf{I} + \delta_k \boldsymbol{\Sigma}_{(m)(C_m)} \mathbf{P}_m (\mathbf{D}_m + \delta_k \mathbf{I})^{-1} \mathbf{D}_m^{-1} \mathbf{P}_m^T \boldsymbol{\Sigma}_{(C_m)(m)} \end{aligned}$$

Finally, we propose the SparseAEH algorithm as follows:

Algorithm 1 SparseAEH

Initialization:

Initialize $\boldsymbol{\mu}_k, \delta_k, \sigma_k^2, \pi_k$

for $m = 1, \dots, M$ **do**

 Diagonalize and store $\boldsymbol{\Sigma}_{(C_m)(C_m)}^{-1} = \mathbf{P}_m \mathbf{D}_m \mathbf{P}_m^T$

 Compute and store $\boldsymbol{\Sigma}_{(m)(C_m)} \boldsymbol{\Sigma}_{(C_m)(m)}, \mathbf{A} = \boldsymbol{\Sigma}_{(m)(C_m)} \mathbf{P}_m$

end for

EM:

while not converge **do**

$\boldsymbol{\Sigma}'_{(m)} \leftarrow \boldsymbol{\Sigma}'_{(m)|\delta_k=0} + \delta_k \mathbf{I} + \delta_k \boldsymbol{\Sigma}_{(m)(C_m)} \mathbf{P}_m (\mathbf{D}_m + \delta_k \mathbf{I})^{-1} \mathbf{D}_m^{-1} \mathbf{P}_m^T \boldsymbol{\Sigma}_{(C_m)(m)}$

 Diagonalize $\boldsymbol{\Sigma}'_{(m)}$ ▷ For faster computation

 Approximate likelihood using equation (4)

$\Omega_{gk} \leftarrow \frac{\pi_k \mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_k, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I}))}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{y}_g | \boldsymbol{\mu}_j, \sigma_k^2 \cdot (\boldsymbol{\Sigma} + \delta_k \mathbf{I}))}$

$\boldsymbol{\mu}_k \leftarrow \frac{\sum_{g=1}^G \Omega_{gk} \mathbf{y}_g}{\sum_{g=1}^G \Omega_{gk}}$

 Update δ_k, σ_k^2 using (9) and (10)

$\pi_k = \frac{\sum_{g=1}^G \Omega_{gk}}{G}$

end while

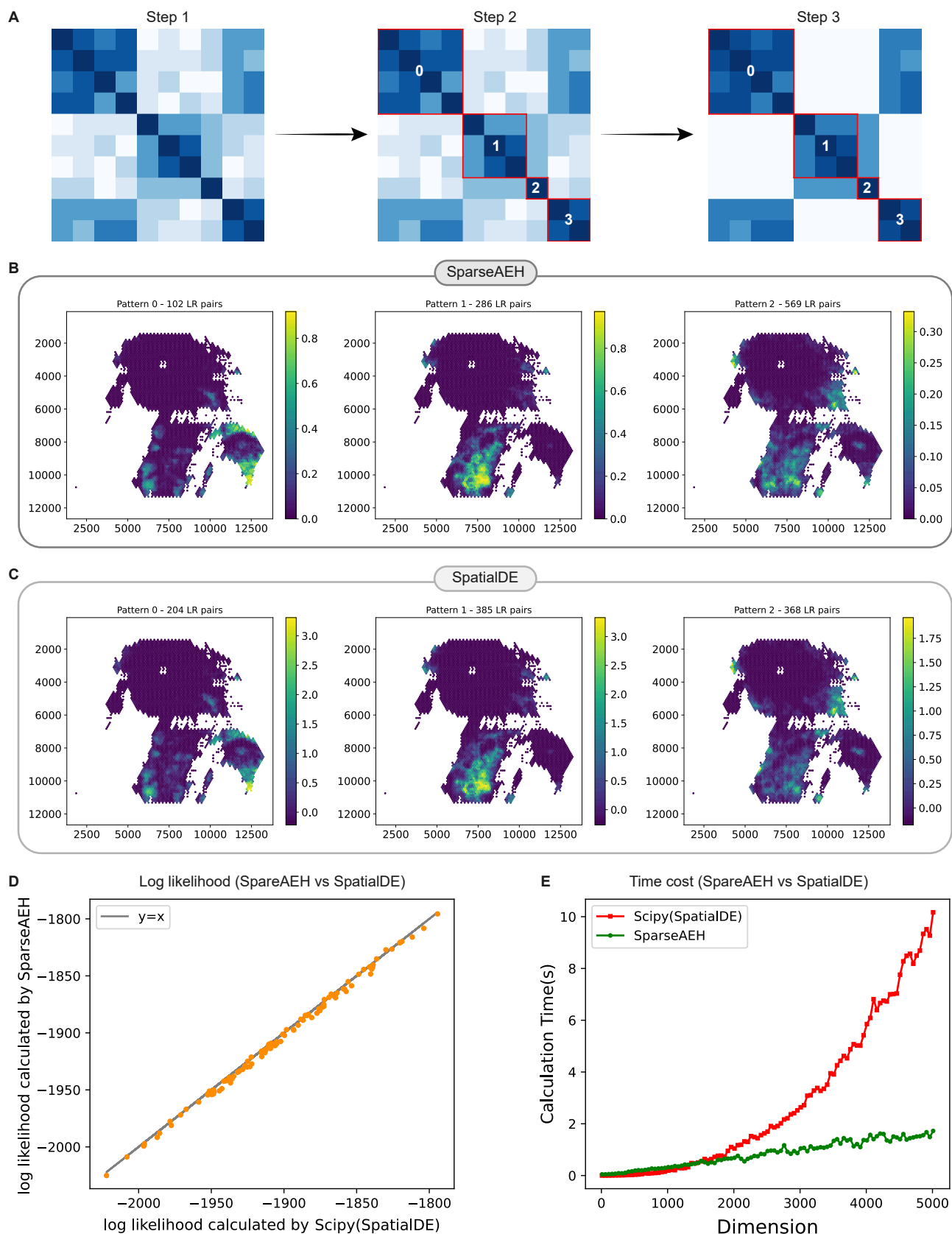


Figure S11. Overview of SparseAEH and its comparison results with SpatialDE in spatial pattern clustering. (A) Typical procedure of SparseAEH, during which the spots are first roughly divided into clusters (B) The clustering results of SparseAEH on the 4× super-resolved NPC dataset from FineST (with 5,039 spots and 957 ligand-receptor pairs), using 12.253 seconds. (C) The clustering results of SpatialDE on the 4× super-resolved NPC dataset from FineST (with 5,039 spots and 957 ligand-receptor pairs), using 21,654.154 seconds. (D) Loglikelihood of simulated data calculated by SparseAEH and Scipy. (E) Time needed to compute likelihood of data of up to 5,000 dimensions.